



Syllabus

EE 471 Modern Software Development Practices and Technologies (3-0)3

Course Information

Course objectives	<i>To provide students with an understanding of the various practical and theoretical concepts of modern software engineering. This will allow them choosing the right career path within the scope of software development.</i>
Course content	<i>Basic coding practices. Software development lifecycle and version control systems. Data types, control flow, functions, object-oriented programming (OOP). Coding Standards and Clean Coding. Cloud Computing Fundamentals. Containerization technologies. Practical AI applications. Front-end development basics. Mobile app development. Back-End Development. Advanced Topics and Industry Trends.</i>
Credits	<i>3 hours/week, 15 weeks in total.</i>
Instructor	<i>Cihan Göksu, PhD.</i>
Lectures times	<i>Monday 13:30 – 16:15 (Classroom Z-40)</i>
Office hours	<i>Monday 16:30 – 17:30</i>
Lecture links	
Course Materials	<i>Given as hyperlinks in the schedule below.</i>
Grading	<i>[20%] In-class exercises [40%] 2 Midterm (4 Mini Projects) [40%] 1 Final Project</i>
Exam dates (tentative)	<i>(Project1) 12/10/2026 Monday; (Project2) 02/11/2026 Monday; (Project3) 23/11/2026 Monday; (Project4) 14/12/2026 Monday; (Final Project) 11/01/2026 Monday;</i>
PREREQUISITE	- Intermediate Python Knowledge - Reliable Hardware (at least 4GB GPU is recommended)

Course Schedule

Week	Topics
#1	Getting Started with Software Engineering - A brain teaser puzzle: You will all conform - Software development Lifecycle and version control system onboarding
#2	Version control systems - An introduction to Gitlab and Github - Basic Git commands and management
#3	Foundational Programming Concepts - Object oriented programming in Python - OOP in Python cont'd
#4	Coding Standards and Clean Coding: - Clean coding and Lint - Vibe coding practices: Cursor-AI, Codex, Chatbots
#5	Introduction to Cloud: Azure AI services - Text to speech example, speech-to-text, real-time diarization - Speech translation, pronunciation-assess, custom-speech (speech studio)
#6	Containerization technologies - Docker: Images, containers, configuration: Practical exercises - Container for AI applications: Replicate/COG
#7	Practical AI applications - Getting started with Pytorch: Quickstart, Tensors, Datasets & DataLoaders, Transforms, Build neural network, optimize the model, save&load - Train a small neural network to classify images
#8	Practical AI applications cont'd: MediaPipe - Vision Tasks
#9	Introduction to Generative AI: A simple Chatbot and artificial content with stable diffusion - Transformers and sentiment analysis - Using transformers, Image generation via Flux
#10	Mobile App and Backend Development Basics - Introduction to Flutter: Develop your first app - Introduction to Flask and Django
#11	Continuous Integration, Continuous Deployment Basics - Building a Github CI/CD Pipeline
#12	Introduction to Agentic AI in Practice - OpenClaw, Hermes Agent, and Crew AI
#13-15	Final Project (Proposals): 3-mins presentation that you describe the problem you are solving and your solution approach. There will be 2-mins of discussion where you'll get feedback.
Finals Week	Final Project (Presentations):5-mins + 3-mins discussion and a max-2-pages report for your project (including your git repo with your codes).